

# Recording for Eyes, Not Echoing to Ears: Contextualized Spoken-to-Written Conversion of ASR Transcripts

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## Abstract

Automatic Speech Recognition (ASR) transcripts exhibit recognition errors and various spoken language phenomena such as disfluencies, ungrammatical sentences, and incomplete sentences, hence suffering from poor readability. To improve readability, we propose a **Contextualized Spoken-to-Written conversion (CoS2W)** task to address ASR and grammar errors and also transfer the *informal* text into the *formal* style with content preserved, utilizing contexts and auxiliary information. This task naturally matches the in-context learning capabilities of Large Language Models (LLMs). To facilitate comprehensive comparisons of various LLMs, we construct a document-level Spoken-to-Written conversion of ASR Transcripts Benchmark (**SWAB**) dataset. Using SWAB, we study the impact of different granularity levels on the CoS2W performance, and propose methods to exploit contexts and auxiliary information to enhance the outputs. Experimental results reveal that LLMs have the potential to excel in the CoS2W task, particularly in grammaticality and formality, our methods achieve effective understanding of contexts and auxiliary information by LLMs. We further investigate the effectiveness of using LLMs as evaluators and find that LLM evaluators show strong correlations with human evaluations on rankings of faithfulness and formality, which validates the reliability of LLM evaluators for the CoS2W task.

## Introduction

As can be seen in Figure 1, since ASR transcripts aim to provide verbatim transcriptions of oral communications, they often exhibit various spoken language phenomena and informal styles, such as filler words, repetitions, repairs, and fragments, and include ASR errors and ungrammatical text. These characteristics lead to poor readability. To enhance the readability of ASR transcripts, we propose the **Contextualized Spoken-to-Written conversion (CoS2W)** task, which aims to correct ASR errors and grammatical errors and transfer the informal style to the written and formal style while preserving the content, that is, the CoS2W task is designed to convert verbatim transcripts (“echoing to ears”) into readable documents (“recording for eyes”). Note that different domains vary in their requirements for adjustments on ASR transcripts. Some domains only allow removal of filler words while some domains (e.g., speech analysis for diagnosis) require verbatim transcripts. Nonetheless, the readable

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|                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Podcast Introduction</b> (auxiliary)                                                                                                                                                                                                                     |
| Discussing Nolan’s movie Oppenheimer with <b>Kerry</b> Lee and <b>Gill</b> Wang ...                                                                                                                                                                         |
| <b>ASR Transcript</b> (source)                                                                                                                                                                                                                              |
| Hello Ms. <b>Carrie</b> . I am <b>Jill</b> . What topic are we discussing today? Everyone, seeing the title, you already know. A blockbuster movie. Nolan Oppenheimer. Yes, aha, Oppenheimer, it feels. This movie <b>has achieve</b> very high box office. |
| <b>CoS2W Record</b> (target)                                                                                                                                                                                                                                |
| Hello Ms. <b>Kerry</b> . I’m <b>Gill</b> . As everyone can tell from the title, today we will discuss Nolan’s blockbuster movie — “Oppenheimer”, which <b>has achieved</b> very high box office revenue.                                                    |

Figure 1: An example of the proposed CoS2W task. CoS2W involves **Text Style Transfer** across sentences, **ASR Error Correction** and **Grammar Error Correction (GEC)**.

documents of CoS2W task are particularly helpful for many domains such as podcasts, education, and meetings (business, project, etc.), where efficiency in information delivery and knowledge acquisition is essential.

Moreover, for a wide variety of downstream tasks on ASR transcripts, such as machine translation, summarization, question-answering (Gupta et al. 2021), and sentiment classification (Zhang et al. 2023), we find that even competitive models (including LLMs) perform substantially better on written text than on ASR transcripts, and the performance on the downstream tasks could benefit substantially from applying CoS2W to ASR transcripts. For example, for machine translation of ASR transcripts, we employ the competitive CSANMT model<sup>1</sup> on the development set of the speech translation BSTC dataset (Zhang et al. 2021b) and find that compared to directly translating ASR transcripts, applying CoS2W (with GPT-4) on ASR transcripts then translating improves the BLEURT (Sellam, Das, and Parikh 2020) score from 57.07 to 61.9.

Although the CoS2W task could improve both readability and performance of downstream tasks of ASR transcripts, there is a lack of research efforts on this task due to its significant challenges, including *task complexity*, *effective exploitation of contexts*, *understanding the impact of granularity*, *the challenge in evaluations*, and *scarcity of labeled data*, which we will detail below.

**Task complexity.** CoS2W combines many subtasks such as ASR error correction, Grammatical Error Correction (GEC), and text style transfer. Specifically, text style transfer focuses

<sup>1</sup><https://www.huggingface.co/modelscope-unofficial/damo-csanmt-zh-en-large>

on converting *informal* spoken language expressions (such as repetitions and redundancies) to the formal style. These subtasks are not simply pipelined but are interconnected: all subtasks are based on a complete semantic understanding of the input and have dependencies among them. ASR and grammar errors can alter semantics, thereby affecting the performance of text style transfer. On the other hand, text style transfer could benefit GEC as GEC performs better on formal texts than spoken texts.

**Understanding the impact of granularity.** Traditional text style transfer focuses on sentence-level conversion. Conversion at a lower granularity (e.g., paragraph-level) instead of sentence-level, may achieve larger improvements to readability, as shown in Figure 1.

However, reducing granularity may introduce faithfulness issues (i.e., divergence from the meaning of the input) as discussed in the results section.

**Effective exploitation of contexts.** We define CoS2W as contextualized due to the critical role of contexts in understanding speech transcripts. For example, context is critical to grasp the meanings of sentence fragments in domains with frequent speaker interactions, such as meetings. Some domains also provide auxiliary information for videos and audios, such as introductions of podcasts. The auxiliary information could benefit ASR error correction as it may contain relevant named entities, domain terms, and words with challenging pronunciations. This information also facilitates understanding of background knowledge and topics, hence helping the informal-to-formal style transfer. Essentially, the auxiliary information can be considered as **extended context**. Effectively leveraging auxiliary information to enhance the CoS2W performance poses a challenge.

**Challenge in evaluations.** Evaluating the CoS2W task is challenging due to its multifaceted nature, requiring different evaluation methods for its subtasks. Moreover, tasks such as informal-to-formal style transfer remain highly subjective and predominantly depend on human evaluations despite the existence of objective metrics. LLMs offer a promising opportunity for automatic evaluations. However, leveraging LLMs as reliable evaluators for CoS2W poses many challenges, including designing appropriate prompts, mitigating bias, and ensuring the accuracy, fairness, and reproducibility of the LLM evaluators.

**Scarcity of labeled data.** Lack of labeled data for CoS2W hinders both model evaluations and efforts to tackle the challenges discussed above.

The advent of LLMs offers new prospects for addressing the challenges of CoS2W. Firstly, the contextualized CoS2W task matches well with LLMs’ in-context learning abilities. Secondly, the CoS2W task considers an output satisfactory as long as it is free of ASR and grammar errors, faithful to the input in semantics, and formal (i.e., close to written text). The generative nature of the problem definition also aligns with LLMs. Thirdly, LLMs have been explored as evaluators for many tasks (Li et al. 2024; Lai, Toral, and Nissim 2023) and achieved notable performance.

Our contributions can be summarized as follows:

- We propose the Contextualized Spoken-to-Written conversion (CoS2W) task to improve both readability of ASR

transcripts and performance of downstream tasks. To promote research in this field, we construct and make available the *document-level Spoken2Written of ASR transcripts Benchmark (SWAB)* dataset with manual annotations, covering meeting, podcast, and lecture domains in both Chinese and English languages.

- We investigate various methods for LLMs to utilize contexts and auxiliary information for CoS2W and provide insights into future research directions.
- We find that LLM evaluators show strong correlations with human evaluations on rankings of faithfulness and formality and also analyze their strengths and weaknesses as CoS2W evaluators.

## Related Work

**ASR Error Correction** task aims to correct misrecognitions within ASR transcripts. Here we briefly summarize the applications of LLMs and auxiliary information.

Recent studies (Min and Wang 2023; Yang et al. 2023) have investigated LLMs’ effectiveness in ASR error correction, using varied prompt strategies like few-shot and Chain of Thought (CoT). While LLMs show potential in this task, their relatively free generation paradigm often results in unnecessary changes beyond fixing ASR errors, such as paraphrasing error-free text. This behavior affects their performance on metrics like Word Error Rates (WER). This problem would also be an issue for the CoS2W task.

In ASR error correction, auxiliary information helps resolve challenging pronunciations, such as N-best transcripts (Ma et al. 2023), dialogue history (Mai and Carson-Berndsen 2023), video titles and descriptions (Lakomkin et al. 2024), rare words (He, Yang, and Toda 2023), etc. However, this auxiliary information is often brief, consisting of just a few sentences or word lists. In addition, as far as we know, studies combining auxiliary information with LLMs in this task are relatively scarce.

**Grammar Error Correction** task entails correcting textual grammatical mistakes. Similar to ASR error correction, LLM performance in GEC isn’t fully satisfactory (Zhang et al. 2023; Fang et al. 2023). GEC datasets usually follow minimum change principle, clashing with LLMs’ relatively free generation paradigm. Disregarding objective metrics, human evaluations confirm LLMs’ GEC effectiveness (Li et al. 2023), which suggests current metrics don’t evaluate LLMs fairly and reliably. Therefore, Li et al. (2024) aim to refine GEC evaluation methods, using LLMs to categorize edits and calculate metrics like  $F_{0.5}$  to evaluate models.

Most GEC task corpora are sentence-level, sourced from second-language learners, and categorized into error-coded and direct rewriting paradigms. Table 1 lists common datasets: For English, FCE (Yannakoudakis, Briscoe, and Medlock 2011) and AESW (Daudaravicius et al. 2016) use error-coded, while JFLEG (Napoles, Sakaguchi, and Tetreault 2017) and WI-LOCNESS (Bryant et al. 2019) use direct rewriting. Chinese datasets include NLPCC18 and Lang-8 (Zhao et al. 2018), CGED (Rao, Yang, and Zhang 2020) from HSK essays, and re-annotated YACLC (Wang et al. 2021) and MuCGEC datasets (Zhang et al. 2022b).

| Dataset      | #Sentences | Granularity     | Annotation | Auxiliary | Language | Domain                       |
|--------------|------------|-----------------|------------|-----------|----------|------------------------------|
| FCE          | 34.0K      | sentence-level  | ✓          | ×         | EN       | FCE Exam Essay               |
| AESW         | 1489.2K    | document-level  | ×          | ×         | EN       | Journal Articles             |
| JFLEG        | 1.5K       | sentence-level  | ✓          | ×         | EN       | TOFEL Exam                   |
| WI-LOCNESS   | 43.1K      | sentence-level  | ✓          | ×         | EN       | Website, Essay               |
| NLPCC18      | 2.0K       | sentence-level  | ✓          | ×         | CH       | Essay                        |
| Lang-8       | 717.0K     | sentence-level  | ×          | ×         | CH       | Language-learning Website    |
| CGED         | 7.2K       | paragraph-level | ✓          | ×         | CH       | HSK Exam                     |
| YACLC        | 32.1K      | sentence-level  | ✓          | ×         | CH       | Website                      |
| MuCGEC       | 7.1K       | sentence-level  | ✓          | ×         | CH       | Essay, HSK Exam, Website     |
| Japanese S2W | 18.2K      | sentence-level  | ✓          | ×         | JA       | Conversation, Voicemail      |
| CS2W         | 7.2K       | sentence-level  | ✓          | ×         | CH       | Telephone Conversation       |
| SWAB         | 29.5K      | document-level  | ✓          | ✓         | EN, CH   | Podcasts, Meetings, Lectures |

Table 1: Comparison between SWAB and other datasets.

**Text Style Transfer** task aims to modify text to a specific style (e.g., informal to formal, negative to positive) while preserving content. The CoS2W task focuses on informal to formal transfer. Recent studies (Reif et al. 2022; Tao et al. 2024) have employed LLMs to improve text style transfer quality and achieve diverse style conversions. In addition, Lai, Toral, and Nissim (2023) employ the LLM as a multi-dimensional evaluator and find that it achieves competitive correlations with human evaluation compared to existing automatic metrics, especially in terms of content preservation. **Spoken-to-Written** task transforms ASR transcripts into formal and readable text, which was initially proposed for Japanese (Ihori, Takashima, and Masumura 2020) and has later been extended to Chinese (Guo et al. 2023).

Different from previous work, the CoS2W task emphasizes the contextualized ability, aiming to convert whole paragraphs across multiple sentences rather than single sentences, with the help of context and auxiliary information. It will further enhance the readability of the results, as illustrated in Figure 1. And it aligns well with the in-context learning capabilities of LLMs. To support this research, we construct the SWAB dataset and report the experiment and evaluation results of LLMs. Additionally, our research spans multiple scenarios and languages as shown in Table 1.

### The SWAB Dataset

To conduct experiments and evaluations of the CoS2W task, we construct the **Spoken2Written of ASR Transcripts Benchmark (SWAB)** dataset<sup>2</sup>, including multiple scenarios (i.e., podcasts, meetings, lectures) in both Chinese and English. There are 60 transcripts with auxiliary information, with each subcategory comprising 10 documents. It is important to note that the SWAB is **document-level** with complete annotations for entire transcripts, providing an annotated target as a reference for each paragraph. Furthermore, we provide links to the original audio or video along with timestamps to support multi-modal research.

### Data Source

We collect Chinese and English data to compare the multilingual capabilities of LLMs, by leveraging the online resources available and selected open-source corpora of spo-

ken transcripts. Additionally, we select three typical scenarios (i.e., meeting, podcast, and lecture) where efficiency in information delivery and knowledge acquisition is essential. Among the three scenarios, meetings are high-frequency interactive discussions among multiple participants; podcasts typically involve chats and interviews between two or more individuals; and lectures are monologues from a single person. Essentially, these three domains exhibit significant differences in interactivity, number of participants, length, etc, hence differing from written texts to varying extents and in turn posing diverse levels of challenges to the CoS2W task.

**Meetings** are sourced from an open-source meeting corpus with manually annotated information as auxiliary information. Chinese meetings are sampled from the training dataset of the AliMeeting corpus (Yu et al. 2022). We select the title and manually annotated topic titles (i.e., sub-topics) as auxiliary information. English Meetings are sampled from the AMI corpus (Carletta et al. 2005). We choose third-party annotated abstractive summarization as auxiliary information, which includes sections on decisions, action items, etc.

**Podcasts** are derived from many Chinese and English podcast programs covering a multitude of topics on YouTube<sup>3</sup>. We collect the podcast introduction as auxiliary information, which provides rich background knowledge such as guest names, specialized terminology, and discussion topics.

**Lectures** are sourced from many Chinese and English individual speeches on YouTube, covering a variety of topics. Both Chinese and English lectures are provided with meta-information of the speeches as auxiliary information, which includes the name and biography of the speaker, as well as the title, category, and description of the video.

### Dataset Construction

We design a relatively free annotation paradigm like text style transfer and direct rewriting of GEC. Targets are considered “correct” as long as they fix ASR and grammatical errors, and adopt a written and formal style while faithfully preserving the original content. In addition, we focus on maintaining consistency of paragraph boundaries between source and target of CoS2W, which facilitates flexible division of content across the document and fair comparison at different granularity levels.

<sup>2</sup><https://www.github.com/alibaba-damo-academy/SpokenNLP/tree/main/swab>

<sup>3</sup><https://www.youtube.com>